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Hybrid solar/wind power system probabilistic modelling for long-term performance assessment

G. Tina *, S. Gagliano, S. Raiti

D. I. E. E. S. University of Catania, v. le A. Doria no. 6, 95125 Catania, Italy

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Abstract

This paper presents a probabilistic approach based on the convolution technique to assess the long-term performance of a hybrid solar–wind power system (HSWPS) for both stand-alone and grid-linked applications. To estimate energy performance of HSWPS the reliability analysis is performed by the use of the energy index of reliability (EIR) directly related to energy expected not supplied (**EENS**). Analytical expressions are developed to obtain the power generated. The hybrid system and the load models employed enable the study period to range from one year to one particular hour-of-day, thus allowing the inclusion of the time-value of energy as appropriate in economic assessments. A numerical example application is included to illustrate the validity of the developed probabilistic model: the results are compared to those resulting from time domain simulations.
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1. Introduction

A growing interest in renewable energy resources has been observed for several years, due to their pollution free, availability in all over the world, and continuity. These facts make these energy resources attractive for many applications. In some cases, however a single energy source system, e.g., a stand alone solar energy system, cannot provide a continuous source of energy due to the low availability during the no-sun period and the winter.

This implies that in order to achieve the high energy availability required in some application such as: lighting, electrification of remote areas and telecommunications it is necessary to oversize the rating of the generating system (e.g., surface of the photovoltaic array, rating of wind turbine). On the other hand it is possible to use hybrid system where two or more renewable energy sources are exploited. In the past the hybrid systems have been considered as preferred for remote systems like radio telecommunication, satellite earth stations, or at sites that are far away from a conventional power system (Godley and Van Lieshout, 1987; Bishop, 1988). Today there is a trend to update the existing one source system (PV, Wind, hydro) into hybrid system also for grid linked application (Bakos and Tsagas, 2003).

Of course the sizing of such systems is much more complicated then single source generating systems, due to the high number of variables and parameters that

* Corresponding author. Tel.: +39 095 738 2337; fax: +39 095 330 793.
E-mail address: gtina@diees.unict.it (G. Tina).

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Nomenclature

A_C	array surface area (m^2)
a_m	number of days in the mth month
$\boldsymbol{C}$	parameter in the probability density function for the hourly clearness index
EENS	energy expected not supplied (kW h)
$EIR_{m,j}$	energy index of reliability
EIR_{ave}	yearly average of EIR_m
EIR_m	monthly average daily EIR
EIR_{max}	maximum $\mathbf{EIR_m}$
EIR_{min}	minimum $\mathbf{EIR_m}$
EIR_y	yearly EIR
E_{year}	average electrical energy produced in a year by HSWPS (kW h)
ESH	equivalent sun hours (h)
$f_{k_t}(k_t)$	probability density function for the hourly clearness index
$f_{P_h}(P_h)$	probability density function for the power output of HSWPS
$F_{P_h}(P_h)$	distribution function for the power output of HSWPS
$f_{P_{pv}}(P_{pv})$	probability density function for the power output of PVS
$F_{P_{pv}}(P_{pv})$	distribution function for the power output of PVS
$f_{P_w}(P_w)$	probability density function for the power output of WECS
$F_{P_w}(P_w)$	distribution function for the power output of WECS
$f_v(v)$	wind speed probability density function
$F_v(v)$	wind speed distribution function
H_o	extra-terrestrial radiation on a horizontal surface (MJ m^{-2})
H_t	global radiation on a horizontal surface (MJ m^{-2})
I_o	extra-terrestrial irradiance on a horizontal surface (kW m^{-2})
I_t	global irradiance on a horizontal surface (kW m^{-2})
I_β	irradiance on a surface with surface tilt angle β to the horizontal plane (kW m^{-2})
k	hourly diffuse fraction
k_t	hourly clearness index
K_t	daily clearness index
k_{tu}	upper bound for k_t
L	local load demand (kW h)
L_{max}	maximum load demand (kW h)
L_{min}	minimum load demand (kW h)
n	day of the year
p	parameter in the functional form for the $k(k_t)$
P_h	power output of HSWPS (kW)
P_{hmax}	maximum power generated by the HSWPS (kW)
P_{hmin}	minimum power generated by the HSWPS (kW)
P_{pv}	power output of PVS (kW)
$P_{pv}(k_{tu})$	maximum power available by the PVS (kW)
P_R	wind turbine rated power (kW)
$P_w(v)$	power output of WECS (kW)
q	parameter in the functional form for the $k(k_t)$
R_b	geometric factor, the ratio of beam radiation on the tilted surface to that on a horizontal surface at any time
r_d	ratio, diffuse radiation in hour/diffuse in day
T and T'	parameters in the functional form for $I_\beta(k_t)$ (kW m^{-2})
$\overline{v}$	average wind speed (m s^{-1})
v	wind speed (m s^{-1})
V_C	cut-in wind speed for the WECS (m s^{-1})
V_F	cut-out wind speed for the WECS (m s^{-1})
V_R	rated wind speed for the WECS (m s^{-1})
α, α'	coefficients of $f_{P_{pv}}(P_{pv})$
α_w	scale parameter Weibull distribution (m s^{-1})
β	tilt angle (°)
β_w	shape parameter Weibull distribution
δ	declination, that is the angular position of the sun at solar noon with respect to the plane of the Equator, north positive
η	efficiency of PVS in realistic reporting conditions (RRC)
λ	parameter in the probability density function for the hourly clearness index
ρ	reflectance of the ground
Φ	latitude, that is the angular location north or south of the equator, north positive ($-90° \leqslant \varphi \leqslant 90°$)
ω	hour angle, that is the angular displacement of the sun east or west of the local meridian due to rotation of the earth on its axis at 15° per hour, morning negative, afternoon positive (°)
ω_s	sunset angle for horizontal surfaces (°)

have to be considered for the optimal design. This type of optimization includes economical objectives, and it requires the assessment of system long-term performance in order to reach the best compromise for both reliability and cost. A reliability assessment of a power system involves evaluation of certain indices which reflect the random nature of system parameters and which serve as reliability measures.

Power system reliability indices can be calculated using a variety of methods. The two main approaches are analytical and simulation methods:

- *Analytical methods* represent the system by means of a mathematical model and evaluate the reliability indices from this model using direct solutions.
- *Simulation methods* estimate the reliability indices by simulating the actual process of the system. The method therefore treats the problem as a series of real experiments. There are several types of simulation processes, but they are all frequently and loosely referred to as Monte Carlo Simulations (MCS). As far as the treatment of input of generating systems based on random variables such as wind speed and solar irradiance and load, three modeling approaches are noted in the literature. The first uses straightforward *chronological temporal series* which require the processing of a large amount of data that may not be available, especially during the design phase (Beyer and Langer, 1996; Giraud and Salameh, 2001). The second approach is based on *probabilistic methods* which treat all the variables participating in the energy conversion process as random variables, and hence performance assessment is achieved by working with distribution functions representing these variables. The last method is the *energy balance* (Markvart, 1996) whose average values (e.g., monthly or daily) are used for simulating the characteristics of input variables (wind energy, solar energy and energy demand). The inefficiency of the model lies in the fact that it gives no insight into the variability of the energy sources and load demand throughout the day.

The second approach has been developed almost exclusively for monovalent system. However, to determine the performance of hybrid system with energy storage, Karaki et al. (1999) have developed a general numerical probabilistic model, the procedure is adapted to determine a solar park model and a wind farm model considering the capacity levels due to hardware failure of the solar modules and wind turbines, the combination of the two models to obtain a hybrid HSWPS is carried using convolution.

The Conti et al. (2000) have developed a probabilistic model and have calculated the long-term system performance in terms of the monthly average fraction of the load met by generation system.

The aim of this paper is to develop an analytical model of the HSWPS using statistical approach and in particular convolving the probability density function (pdf) of power generated by solar and wind generating systems. This approach overcomes the error in the convolution calculation introduced in the model developed by Karaki et al. due to the discretization of the probability density function of wind and solar power output. In this paper the reliability indices used is the energy index of reliability EIR, directly related to expected energy not supplied EENS, which is normalized by dividing by the total energy load demand. To show the validity of the proposed model, a numerical example is also presented; the value of EIR, evaluated by means of the closed form expression reported in the paper, is compared with the one resulting from time domain simulations.

2. System description and models

In the system under study a wind energy conversion system (WECS) is connected to the control unit through an appropriate electrical interface. In parallel a photovoltaic system (PVS) is also connected to the control unit. Interaction with the grid is assumed to be bi-directional: excess of energy generated by the HSWPS is conditionally supplied to the grid. On the other hand, deficit of energy will be drawn by the grid in the low generating phase to supply the local demand (see Fig. 1).

The performance of the HSWPS under study can be assessed by employing suitable probabilistic models for both the WECS and PVS.

2.1. WECS model

The wind speed distribution for selected sites as well as the power output characteristic of the chosen wind turbine are the factors that have to be considered to determine the WECS power output.

Since the wind speed "v" is a random variable, a long-term meteorological data is desirable to describe wind energy potential of the sites. In order to account the variability of wind speed, during the jth hour ($j = 1, \ldots, 24$) of the mth month ($m = 1, \ldots, 12$), it is assumed to be characterized by a Weibull distribution with a scale parameter α_w and a shape parameter β_w. Density and distribution probability functions are given by

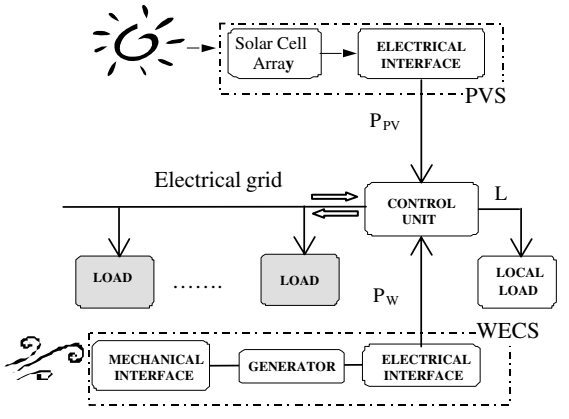


Fig. 1. Schematic of the HSWPS connected to the grid.

(hereafter for legibility's sake the variables that depend on the hour of the day and on the month will be written in boldface):

$$\boldsymbol{f}_{\boldsymbol{v}}(\boldsymbol{v}) = \frac{\boldsymbol{\beta}_{\mathrm{w}}}{\boldsymbol{\alpha}_{\mathrm{w}}^{\boldsymbol{\beta}_{\mathrm{w}}}} \cdot \boldsymbol{v}^{\boldsymbol{\beta}_{\mathrm{w}}-1} \cdot \exp\left[-\left(\frac{\boldsymbol{v}}{\boldsymbol{\alpha}_{\mathrm{w}}}\right)^{\boldsymbol{\beta}_{\mathrm{w}}}\right] \tag{1}$$

$$\boldsymbol{F}_{\boldsymbol{v}}(\boldsymbol{v}) = 1 - \exp\left[-\left(\frac{\boldsymbol{v}}{\boldsymbol{\alpha}_{\mathrm{w}}}\right)^{\boldsymbol{\beta}_{\mathrm{w}}}\right] \tag{2}$$

For a typical WECS, the power output characteristic can be assumed in such a way that it starts generating at the cut-in wind speed V_{C}, the power output increases linearly as the wind speed increases from V_{C} to the rated wind speed V_{R}. The rated power P_{R} is produced when the wind speed varies from V_{R} to the cut-out wind speed V_{F}. at which the WECS will be shut down for safety.

Thus,

$$\boldsymbol{P}_{\mathrm{w}}(\boldsymbol{v}) = \begin{cases} \left(\frac{P_{\mathrm{R}}}{V_{\mathrm{R}}-V_{\mathrm{C}}}\right) \cdot (\boldsymbol{v} - V_{\mathrm{C}}) & \text{for } V_{\mathrm{C}} \leqslant \boldsymbol{v} \leqslant V_{\mathrm{R}} \\ P_{\mathrm{R}} & \text{for } V_{\mathrm{R}} \leqslant \boldsymbol{v} \leqslant V_{\mathrm{F}} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The probability density function $\boldsymbol{f}_{\boldsymbol{P}_{\mathrm{w}}}(\boldsymbol{P}_{\mathrm{w}})$ for the power output of the WECS can be obtained using Eqs. (1) and (3) by the application of the transformation theorem (Abouzahr and Ramakumar, 1991) as

$$\boldsymbol{f}_{\boldsymbol{P}_{\mathrm{w}}}(\boldsymbol{P}_{\mathrm{w}}) = \begin{cases} \boldsymbol{F}_1 & \boldsymbol{P}_{\mathrm{w}} = 0 \\ \left(\frac{V_{\mathrm{R}}-V_{\mathrm{C}}}{P_{\mathrm{R}}}\right) \cdot \left(\frac{\boldsymbol{\beta}_{\mathrm{w}}}{\boldsymbol{\alpha}_{\mathrm{w}}^{\boldsymbol{\beta}_{\mathrm{w}}}}\right) & \\ \times \left(V_{\mathrm{C}} + (V_{\mathrm{R}} - V_{\mathrm{C}}) \cdot \frac{\boldsymbol{P}_{\mathrm{w}}}{P_{\mathrm{R}}}\right)^{\boldsymbol{\beta}_{\mathrm{w}}-1} & \\ \times \exp\left[-\left(\frac{V_{\mathrm{C}}+(V_{\mathrm{R}}-V_{\mathrm{C}})\frac{\boldsymbol{P}_{\mathrm{w}}}{P_{\mathrm{R}}}}{\boldsymbol{\alpha}_{\mathrm{w}}}\right)^{\boldsymbol{\beta}_{\mathrm{w}}}\right] & 0 < \boldsymbol{P}_{\mathrm{w}} < P_{\mathrm{R}} \\ \boldsymbol{F}_2 & \boldsymbol{P}_{\mathrm{w}} = P_{\mathrm{R}} \end{cases} \tag{4}$$

where

$$\begin{aligned} \boldsymbol{F}_1 &= 1 - [\boldsymbol{F}_v(V_{\mathrm{F}}) - \boldsymbol{F}_v(V_{\mathrm{C}})] \\ \boldsymbol{F}_2 &= \boldsymbol{F}_v(V_{\mathrm{F}}) - \boldsymbol{F}_v(V_{\mathrm{R}}) \end{aligned} \tag{5}$$

Using Eqs. (4) and (5), the resulting plots of probability density function and distribution function for $\boldsymbol{P}_{\mathrm{w}}$ are shown, respectively, in Fig. 2a and b. In Fig. 2a the two arrows indicate the two impulses referring respectively to $\boldsymbol{f}_{\boldsymbol{P}_{\mathrm{w}}}(0)$ and $\boldsymbol{f}_{\boldsymbol{P}_{\mathrm{w}}}(P_{\mathrm{R}})$ (see Fig. 2a and b).

2.2. PVS model

The amount of solar radiation that reaches the ground, besides on the daily and yearly apparent motion of the sun, depends on the geographical location (latitude and altitude) and on the climatic conditions (e.g., cloud cover). Many studies have proved that cloudiness is the main factor affecting the difference between the values of solar radiation measured outside the atmosphere and on earthly surface. To account for the difference between these two values, a *daily clearness index*, $\boldsymbol{K}_{\mathrm{t}}$, has been defined as the ratio of a particular day's total solar radiation, $\boldsymbol{H}_{\mathrm{t}}$ [MJ m^{-2}], to the extraterrestrial total solar radiation, $\boldsymbol{H}_{\mathrm{o}}$ [MJ m^{-2}], for that day, both referred to a horizontal surface. We can also define an *hourly clearness index*, $\boldsymbol{k}_{\mathrm{t}}$, as the ratio of the irradiance on an horizontal plane, $\boldsymbol{I}_{\mathrm{t}}$ [kW m^{-2}], to the extraterrestrial total solar irradiance $\boldsymbol{I}_{\mathrm{o}}$ [kW m^{-2}]:

$$\boldsymbol{k}_{\mathrm{t}} = \frac{\boldsymbol{I}_{\mathrm{t}}}{\boldsymbol{I}_{\mathrm{o}}} \tag{6}$$

Known the *hourly clearness index* $\boldsymbol{k}_{\mathrm{t}}$ it is possible to determine the *irradiance* on a surface with inclination β to the horizontal plane, $\boldsymbol{I}_{\beta}$ [kW m^{-2}] (see Appendix B). Since the PVS is usually equipped with a maximum power point tracker (MPPT) and the relationship

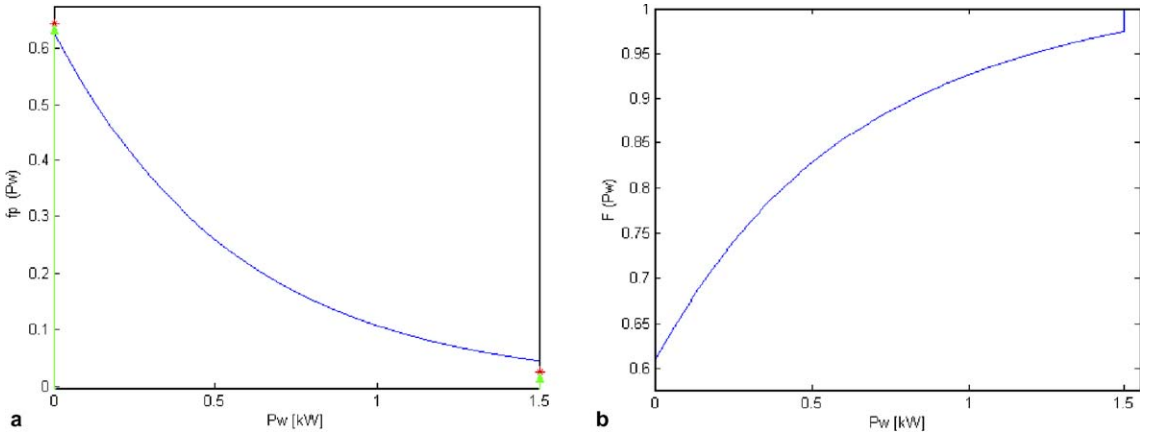


Fig. 2. (a) Probability density function for $\boldsymbol{P}_{\mathrm{w}}$ in October at 9:00 a.m. ($\boldsymbol{\alpha}_{\mathrm{w}} = 4.35$ m s^{-1}, $\boldsymbol{\beta}_{\mathrm{w}} = 1$, $P_{\mathrm{R}} = 1.5$ kW, $V_{\mathrm{C}} = 3.6$ m s^{-1}, $V_{\mathrm{R}} = 12.5$ m s^{-1}, $V_{\mathrm{F}} = 14.5$ m s^{-1}). (b) Cumulative probability function for $\boldsymbol{P}_{\mathrm{w}}$ in October at 9:00 a.m. ($\boldsymbol{\alpha}_{\mathrm{w}} = 4.35$ m s^{-1}, $\boldsymbol{\beta}_{\mathrm{w}} = 1$, $P_{\mathrm{R}} = 1.5$ kW, $V_{\mathrm{C}} = 3.6$ m s^{-1}, $V_{\mathrm{R}} = 12.5$ m/s, $V_{\mathrm{F}} = 14.5$ m s^{-1}).

between the maximum power per unit area of array surface available from PVS and I_β is linear, (Kroposki et al., 1994), the power output of PVS (P_{pv}) is given by

$$P_{pv} = A_C \cdot \eta \cdot I_\beta = A_C \cdot \eta \cdot (T \cdot k_t - T' \cdot k_t^2) \quad (7)$$

where A_C is the array surface area [m^2]; I_β the *irradiance* on a surface with inclination β to the horizontal plane [kW m^{-2}]; η is the efficiency of the PVS in realistic reporting conditions (RRC) (Gay et al., 1982); T and T' are parameters that depend on inclination β, declination δ, reflectance of the ground ρ, latitude φ, hour angle ω, sunset hour angle ω_s, day of the year n (Appendix B).

From (7), if the probability density function $f_{k_t}(k_t)$ for the random variable k_t is known, it is possible to obtain the probability density function $f_{P_{pv}}(P_{pv})$ for P_{pv} by applying the fundamental theorem for function of a random variable (Papoulis, 1982). In the present work the expression used for the probability density function of k_t is the one proposed by Hollands and Huget (Hollands and Huget, 1983).

Depending on the sign of the parameters T and T', the probability density function $f_{P_{pv}}(P_{pv})$ has four different expressions but only two have a physical meaning (Conti et al., 2002).

In particular, if $T > 0$ and $T' < 0$ we have

$$f_{P_{pv}}(P_{pv}) = \begin{cases} \frac{C\cdot\left(k_{tu}-\frac{1}{2}\cdot(\alpha+\alpha')\right)}{-k_{tu}\cdot A_C\cdot\eta\cdot T'\cdot\alpha'}\cdot \ell^{\frac{\lambda}{2}(\alpha+\alpha')} & \text{if } P_{pv}\in[0,P_{pv}(k_{tu})] \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

while, if $T > 0$ and $T' > 0$ we have

$$f_{P_{pv}}(P_{pv}) = \begin{cases} \frac{C\cdot\left(k_{tu}-\frac{1}{2}\cdot(\alpha-\alpha')\right)}{k_{tu}\cdot A_C\cdot\eta\cdot T'\cdot\alpha'}\cdot \ell^{\frac{\lambda}{2}(\alpha-\alpha')} & \text{if } P_{pv}\in[0,P_{pv}(k_{tu})] \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where

$$\alpha = \frac{T}{T'} \quad \text{and} \quad \alpha' = \sqrt{\alpha^2 - 4\cdot\frac{P_{pv}}{\eta\cdot T'\cdot A_C}} \quad (10)$$

k_{tu} is the upper bound for k_t; C, λ are parameters of the probability density function for k_t (Hollands and Huget, 1983).

Typical plots of the probability density function and cumulative function of P_{pv} are shown in Fig. 3a and b.

2.3. Hybrid system model

The power generated by an hybrid system P_h is equal to the power output of the WECS plus the power output of the PVS P_{pv}:

$$P_h = P_w + P_{pv} \quad (11)$$

Assuming that P_w and P_{pv} are statistically independent, the density function for the random variable P_h is found by convolving the density functions of P_w and P_{pv}.

$$f_{P_h}(P_h) = f_{P_w}(P_w) * f_{P_{pv}}(P_{pv}) \quad (12)$$

On the other hand when there is a dependency between solar irradiance and wind speed, it is possible to use (12) by dividing the study period into time-frames of one or two hours.

According to the system's working conditions, the probability density function of P_h and the integration's extremes of the convolution integral of expressions (12) will vary (Appendix A). We have

$$f_{P_h}(P_h) = \int_{\max(0,P_h-P_{pv}(k_{tu}))}^{\min(P_h,P_R)} f_{P_{pv}}(P_h - P_w)\cdot f_{P_w}(P_w)\,dP_w \quad (13)$$

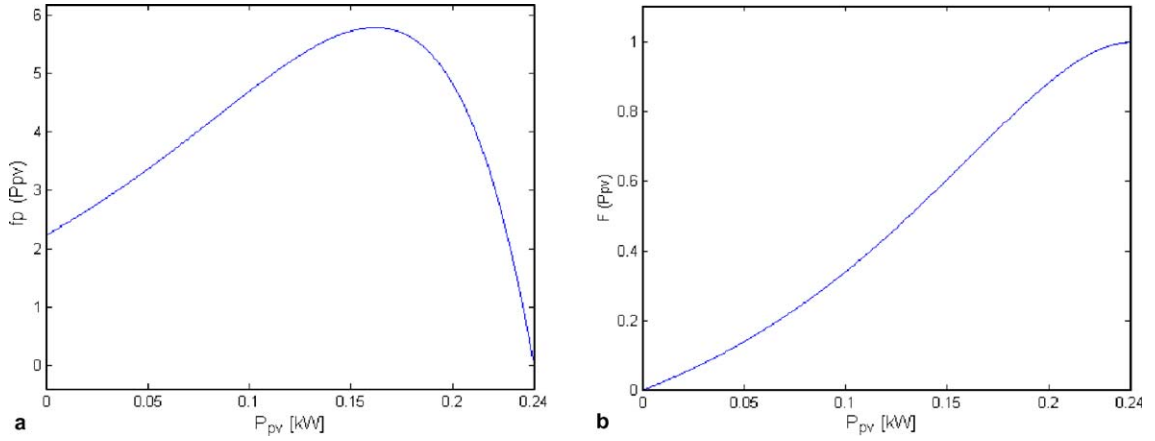


Fig. 3. (a) Probability density function for P_{pv} in October at 9:00 a.m. ($\beta = 0$, $C = 0.6251$, $\lambda = 3.5507$, $k_{tu} = 0.864$, $P_{pv}(k_{tu}) = 0.24$ kW, $A_C = 5$ m^2). (b) Cumulative probability function for P_{pv} in October at 9:00 a.m. ($\beta = 0$, $C = 0.6251$, $\lambda = 3.5507$, $k_{tu} = 0.864$, $P_{pv}(k_{tu}) = 0.24$ kW, $A_C = 5$ m^2).

The power generated by the two subsystems ranges as follow:

$$0 \leqslant \boldsymbol{P}_{\mathrm{pv}} \leqslant \boldsymbol{P}_{\mathrm{pv}}(k_{\mathrm{tu}}) \quad \text{for PVS;}$$
$$0 \leqslant \boldsymbol{P}_{\mathrm{w}} \leqslant P_{\mathrm{R}} \quad \text{for WECS;}$$
$$\boldsymbol{P}_{\mathrm{hmin}} \leqslant \boldsymbol{P}_{\mathrm{h}} \leqslant \boldsymbol{P}_{\mathrm{hmax}} \quad \text{for HSWPS}$$

where $\boldsymbol{P}_{\mathrm{hmax}} = \boldsymbol{P}_{\mathrm{R}} + \boldsymbol{P}_{\mathrm{pv}}(k_{\mathrm{tu}})$ is the maximum power generated by the HSWPS; $\boldsymbol{P}_{\mathrm{hmin}}$ is the minimum power generated by the HSWPS and it is assumed to be 0.

Starting from the analytical expression of the $\boldsymbol{f}_{\boldsymbol{P}_{\mathrm{h}}}(\boldsymbol{P}_{\mathrm{h}})$ a numerical code in Matlab® environment has been developed to calculate the pdf for $\boldsymbol{P}_{\mathrm{h}}$.

Using the results of the proposed model, examples of probability density function and distribution function curves for the power generated by HSWPS are given in Fig. 4a and b.

2.4. Expected energy not supplied (*EENS*)

Expected energy not supplied is a probabilistic index used in reliability analysis for power systems using random renewable energy sources. The **EENS** is the expected energy that will not be supplied due to those occasions when the load exceeds the available generation.

EENS for the HSWPS assumes different forms depending on load and generating conditions.

If the local load $\boldsymbol{L}$ is greater than $\boldsymbol{P}_{\mathrm{hmax}}$ the **EENS** can be easily computed from

$$\mathbf{EENS} = \boldsymbol{L} - \int_{\boldsymbol{P}_{\mathrm{hmin}}}^{\boldsymbol{P}_{\mathrm{hmax}}} \boldsymbol{P}_{\mathrm{h}} \cdot \boldsymbol{f}_{\boldsymbol{P}_{\mathrm{h}}}(\boldsymbol{P}_{\mathrm{h}})\,\mathrm{d}\boldsymbol{P}_{\mathrm{h}} \tag{14}$$

otherwise if $\boldsymbol{P}_{\mathrm{hmin}} \leqslant \boldsymbol{L} \leqslant \boldsymbol{P}_{\mathrm{hmax}}$

$$\mathbf{EENS} = \int_{\boldsymbol{P}_{\mathrm{hmin}}}^{\boldsymbol{L}} (\boldsymbol{L} - \boldsymbol{P}_{\mathrm{h}}) \cdot \boldsymbol{f}_{\boldsymbol{P}_{\mathrm{h}}}(\boldsymbol{P}_{\mathrm{h}})\,\mathrm{d}\boldsymbol{P}_{\mathrm{h}} \tag{15}$$

if $\boldsymbol{L} < \boldsymbol{P}_{\mathrm{hmin}}$

$$\mathbf{EENS} = 0 \tag{16}$$

The expressions of $\boldsymbol{f}_{\boldsymbol{P}_{\mathrm{h}}}(\boldsymbol{P}_{\mathrm{h}})$ vary with the operating conditions (see Appendix A, where, for legibility's sake, the expressions are reported for a particular case: $\beta = 0$).

The energy index of reliability is then given by

$$\mathrm{EIR}_{m,j} = 1 - \frac{\mathbf{EENS}_{m,j}}{L_{m,j}} \quad \text{(on hourly basis)} \tag{17}$$

$$\mathrm{EIR}_m = 1 - \frac{\sum_{j=1}^{24} a_m \cdot \mathbf{EENS}_{m,j}}{\sum_{j=1}^{24} a_m \cdot L_{m,j}} = 1 - \frac{\sum_{j=1}^{24} \mathbf{EENS}_{m,j}}{\sum_{j=1}^{24} L_{m,j}} = \mathrm{EIR}_d$$

(on monthly and daily basis) (18)

$$\mathrm{EIR}_{\mathrm{y}} = 1 - \frac{\sum_{m=1}^{12} \mathbf{EENS}_m}{\sum_{m=1}^{12} L_m} \quad \text{(on yearly basis)} \tag{19}$$

where a_m is the number of days of the mth month.

Billinton and Allan (1996) shows that MCS can be applied to the same type of problem and produce the same results as the analytical approach, and the MCS approach can produce extended sets of results compared with the analytical method.

Simulation models based on Monte Carlo-type algorithms are usually very time-consuming and complex and unsuitable for a rough sizing. They, moreover, require detailed information that is often not available. Finally, simulation algorithms do not make relations existing between design parameters evident and so make more difficult the identification of those variables influencing the system performances most. In order to overcome these limitations, the use of an analytical approach is generally suggested.

To test the accuracy of the numerical results obtained by means of the proposed probabilistic method a MCS have been ran. As long as the numbers of simulations (at least one thousand samples) increases, the Monte

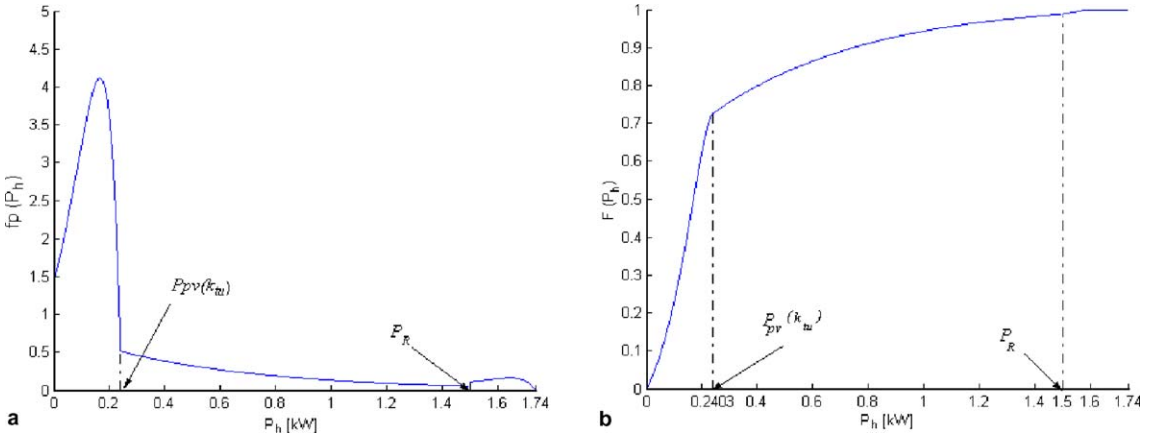


Fig. 4. (a) Probability density function for $\boldsymbol{P}_{\mathrm{h}}$ in October at 9:00 a.m. (b) Cumulative density function for $\boldsymbol{P}_{\mathrm{h}}$ in October at 9:00 a.m.

Carlo results become very close to the probabilistic model.

3. Numerical example

A remote place "Ginostra" (latitude 38.11°, longitude 15.2°) in a little island "Stromboli", in Aeloian Archipelago (Italy), is chosen as a case study for the application of the HSWPS model presented in this paper. From the renewable energy point of view this place is characterized by a low wind speed regime (yearly average wind speed $\bar{v} = 4.55$ m/s) and high values of insulation (equivalent sun hours, ESH = 5.5 h). In Table 1 the monthly average daily pdf parameters of wind and solar regimes are reported for the considered place (see Table 1).

The advantages of combining the use of wind and solar energy for renewable electricity supply systems depend on a seasonal anticorrelation in the time pattern of the wind and solar resources. Fig. 5 shows, for the case study, the annual trends of the monthly electrical energy produced by a PVS (with two different tilt angles of PV array) and by a WECS with the same rated power. It can be observed an evident anticorrelation (see Fig. 5).

The main driving element of any power generating system is the load. A specific feature of Ginostra is the strong variability of the inhabitants' number during a year, about 50 people in winter and approximately 600 in summer, consequently the ratio between the maximum load (L_{max}) and the minimum one (L_{min}) over one year is quite high. In Fig. 6 the hourly monthly average daily load curve in August (the peak load demand) and in January (the low load demand) is reported. It is worth noticing that, in our analysis, the base of time is one hour, so numerically the power in each hour has the same value of the energy in the same hour (e.g., 100 kW for 1 h is 100 kW h) (see Fig. 6).

Table 1
Pdf parameters of wind and solar regimes for the case study

Months	Wind		Solar	
	α_w [m/s]	β_w	C	λ
January	4.921	1.01	0.2900	1.21
February	5.094	1.02	0.2656	1.56
March	4.921	1.01	0.2582	2.22
April	4.921	1.01	0.1787	2.7
May	5.015	1.03	0.1239	3.24
June	3.609	1.1	0.0639	4.51
July	3.893	1.02	0.0434	5.69
August	4.285	1.01	0.0547	4.93
September	3.846	1	0.0992	3.62
October	4.348	1	0.1755	2.38
November	4.762	1	0.2801	1.53
December	4.921	1.01	0.3292	1.1

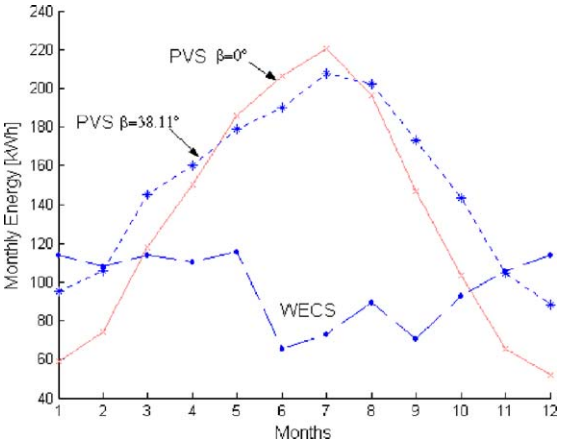


Fig. 5. Monthly electrical energy produced by a 1 kWp PVS, $\eta = 0.093$, tilt angles of PV array $\beta = 0°$ (solid line) and $\beta = 38.11°$ (dotted line), monthly electrical energy produced by a WECS with $P_R = 1$ kW, $V_C = 4.5$ m s^{-1}, $V_R = 16$ m s^{-1}, $V_F = 60$ m s^{-1}.

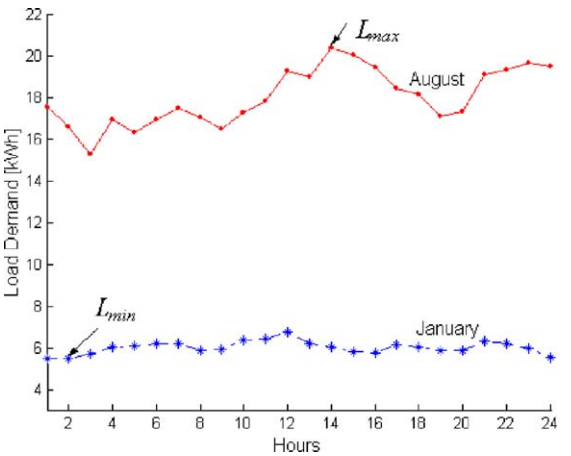


Fig. 6. Hourly monthly average daily load curve in August and in January ($L_{max} = 20.33$ kW h, $L_{min} = 5.454$ kW h, $L_{max}/L_{min} = 3.73$).

For each month of the year the maximum/minimum power of the monthly average daily load demand are reported in Fig. 7 as well the daily energy. As shown in Fig. 7 the maximum load demand is registered in summer when the electrical energy produced by a PVS is maximum. Comparing Figs. 5 and 7 it is possible to note that the maximum load demand and the maximum energy production of PVS are substantially overlapped (see Fig. 7).

In Tables 2 and 3 fixed the rating of the HSWPS, some significant values of EIR (minimum, maximum and yearly average of EIR_m and yearly) by different rating combinations of PVS and WECS are reported. The

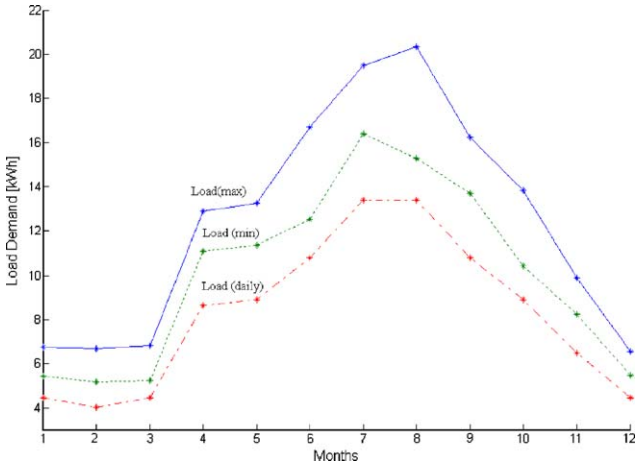


Fig. 7. Hourly monthly average—maximum [kW h] (solid line)/minimum [kW h] (dotted line) load demand/daily $* 10^{-3}$ [kW h] (dashed-dotted line) energy load demand.

Table 2
EIR for an HSWPS varying surface's PVS array and WECS rated power for $\beta = 38.11°$

Case	P_R [kW]	P_{pv} [kW$_p$]	E_{year} [kW h]	EIR_{min}	EIR_{ave}	EIR_{max}	EIR_y
0	0	80	143,520	0.000	0.403	1.000	0.418
1	20	60	131,090	0.100	0.523	1.000	0.519
2	40	40	118,650	0.160	0.540	0.990	0.537
3	60	20	106,180	0.199	0.510	0.989	0.501
4	80	0	93,894	0.222	0.313	0.385	0.293

Table 3
EIR for an HSWPS varying surface's PVS array and WECS rated power for $\beta = 0°$

Case	P_R [kW]	P_{pv} [kW$_p$]	E_{year} [kW h]	EIR_{min}	EIR_{ave}	EIR_{max}	EIR_y
0	0	80	126,270	0.000	0.426	0.999	0.454
1	20	60	118,180	0.100	0.538	0.999	0.544
2	40	40	110,080	0.164	0.550	0.995	0.553
3	60	20	101,990	0.200	0.500	0.970	0.496
4	80	0	93,894	0.222	0.313	0.385	0.293

rated power of HSWPS ($P_h = 80$ kW) has been chosen in such a way that the average electrical energy produced in a year by the various combinations, reported in the tables, is roughly comparable to the yearly load demand (98,800 kW h) (see Tables 2 and 3).

From the comparison between the Tables 2 and 3 is evident that the optimum tilt of PV array is $\beta = 0°$, instead of the fact for the β equal to the site latitude the max production of energy is collected. This fact can be explained by means the plot of the EIR_m (Fig. 8) for two different surface inclinations PV array, $\beta = 0°$ and $\beta = 38.11°$, in fact there is a coupling between PVS generation and load demand: in summer the optimum surface inclination is $\beta = 0°$ (see Fig. 5) and in the same period the load demand is maximum (see Fig. 7).

As shown the results reported in Tables 2 and 3 the correlation between the two energy sources and the load curve is such that the EIR_y with many different HSWPS configurations (cases 1–3) does not vary meaningfully, so to choose the most cost-effective combination it is necessary to solve an optimization problem.

Finally application results of the proposed analytical method have been compared to the outcome of a simulation program, developed in Matlab-Simulink® environment, based on Monte Carlo Simulation (MCS) that has been developed to perform simulation studies of hybrid generation systems (Tina et al., 2004). Fig. 9 shows that the EIR_y obtained by means of MCS method, after about fifty simulation years, converges to the probabilistic model value. In Fig. 10 it is possible to

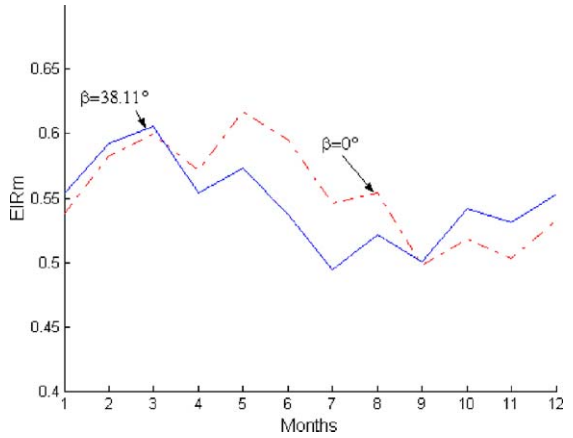


Fig. 8. Monthly average daily EIR_m for HSWPS ($P_R = 40$ kW, $V_C = 4.5$ m s^{-1}, $V_R = 16$ m s^{-1}, $V_F = 60$ m s^{-1}, peak power of PVS = 40 kWp, $\eta = 0.093$).

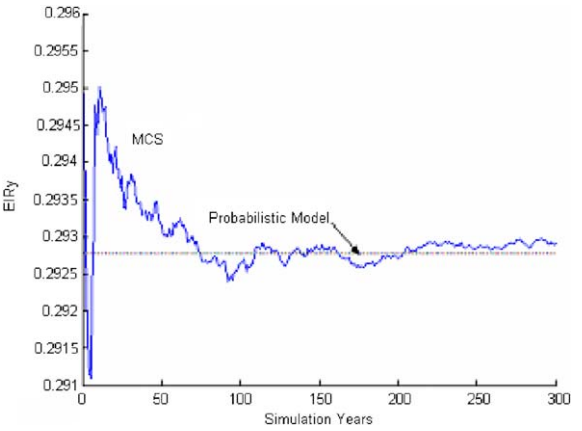


Fig. 9. Comparison between numerical and analytical results for EIR_y ($P_R = 80$ kW, PVS = 0 kWp, $\beta = 38.11°$).

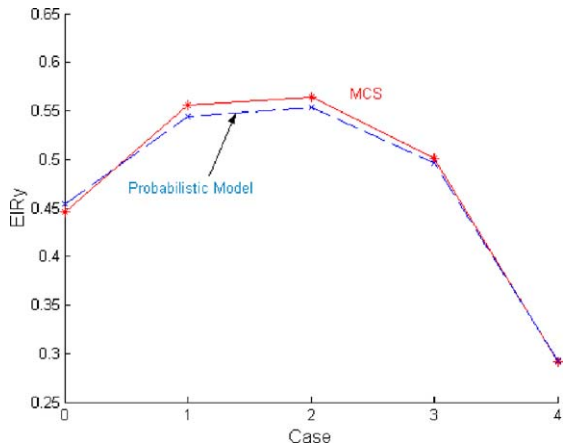


Fig. 10. Comparison between numerical and analytical results for EIR_y referred to the cases reported in Table 3.

observe a comparison between the two methods results for different HSWPS configurations. Percentage difference between analytical and numerical results ranges between 0.5% and 1.9% (see Figs. 9 and 10).

4. Conclusions

In the framework of optimal sizing of hybrid generation systems based on the use of wind and solar energy sources, this paper presents a new probabilistic model that allows to estimate the long-term average performance of a hybrid solar–wind power system. The proposed model is based on a closed form solution approach to convolute the WECS and PVS output power.

This paper has shown by means of a case study, that a good evaluation of the long-term average performance of a hybrid system can be obtained through a statistical approach alternative to time step simulations.

The numerical example presented illustrates the versatility of the approach developed especially in the perspective of the optimal unit sizing of a grid connected HSWPS system without storage batteries, including economical objectives (i.e., electric contract demand, expected values of annual total cost and annual energy consumption).

The results of the analysis can be used not only to calculate the EIR but also to document many other relationships between different system parameters of interest.

Appendix A

Probability density functions ($f_{iP_h}(P_h)$ for $\beta = 0°$

$$f_{iP_h}(P_h) = \int_{L_i}^{U_i} \frac{C \cdot k_{tu} \cdot \left(\frac{V_R - V_C}{P_R}\right) \cdot \left(\frac{\beta_w}{\alpha_w^{\beta_w}}\right)}{P_{pv}(k_{tu})} \cdot \left(V_C + (V_R - V_C) \cdot \frac{P_w}{P_R}\right)^{\beta_w - 1} \cdot \left(1 - \frac{P_h - P_w}{P_{pv}(k_{tu})}\right) \cdot \exp\left[-\left(\frac{V_C + \left(\frac{V_R - V_C}{P_R}\right) \cdot P_w}{\alpha_w}\right)^{\beta_w} + \left(\frac{\lambda \cdot k_{tu}}{P_{pv}(k_{tu})} \cdot (P_h - P_w)\right)\right] dP_w + A + B \tag{A.1}$$

where $U_i = \min(P_h, P_R)$ is the upper integration's extreme; $L_i = \max(0, P_h - P_{pv}(k_{tu}))$ is the lower integration's extreme.

Table A1
Probability density functions varying working conditions

$f_{iP_h}(P_h)$	Working conditions	
$f_{1P_h}(P_h)$	$P_R \leqslant P_{pv}(k_{tu})$	$0 \leqslant P_h < P_R$
$f_{2P_h}(P_h)$		$P_R \leqslant P_h \leqslant P_{pv}(k_{tu})$
$f_{3P_h}(P_h)$		$P_{pv}(k_{tu}) < P_h \leqslant P_{pv}(k_{tu}) + P_R$
$f_{4P_h}(P_h)$	$P_R > P_{pv}(k_{tu})$	$0 \leqslant P_h \leqslant P_{pv}(k_{tu})$
$f_{5P_h}(P_h)$		$P_{pv}(k_{tu}) < P_h < P_R$
$f_{6P_h}(P_h)$		$P_R \leqslant P_h \leqslant P_R + P_{pv}(k_{tu})$

Table A2
Integration's extremes and impulses for $(f_{iP_h}(P_h))$ reported in Table A1 (yes means that the impulse is not equal to zero, whereas NO means that the impulse is null)

	$f_{1P_h}(P_h)$	$f_{2P_h}(P_h)$	$f_{3P_h}(P_h)$	$f_{4P_h}(P_h)$	$f_{5P_h}(P_h)$	$f_{6P_h}(P_h)$
L_i	0	0	$P_h - P_{pv}(k_{tu})$	0	$P_h P_{pv}(k_{tu})$	$P_h - P_{pv}(k_{tu})$
U_i	P_h	P_R	P_R	P_h	P_h	P_R
A	Yes	No	No	Yes	Yes	No
B	No	No	Yes	No	Yes	Yes

$$A = F_1 \cdot C \cdot k_{tu} \cdot \left(1 - \frac{P_h}{P_{pv}(k_{tu})}\right) \cdot \exp\left[\frac{\lambda \cdot k_{tu} \cdot P_h}{P_{pv}(k_{tu})}\right] \tag{A.2}$$

$$B = F_2 \cdot C \cdot k_{tu} \cdot \left(1 - \frac{(P_h - P_R)}{P_{pv}(k_{tu})}\right) \cdot \exp\left[\frac{\lambda \cdot k_{tu} \cdot (P_h - P_R)}{P_{pv}(k_{tu})}\right] \tag{A.3}$$

See Tables A1 and A2.

Appendix B

PVS Probabilistic model (Conti et al., 2002).

The irradiance on a surface with inclination β to the horizontal plane, I_β, is

$$I_\beta = \left[R_b + \left(\frac{1+\cos\beta}{2} - R_b\right) \cdot k + \rho \cdot \frac{1-\cos\beta}{2}\right] \cdot I_t \quad [kW/m^2] \tag{B.1}$$

where R_b is the ratio of beam radiation on the tilted surface to that on a horizontal surface at any time (Duffie and Beckman, 1991); k is the fraction of the hourly radiation on horizontal plane which is diffuse; ρ is the reflectance of the ground.

Further I_t can be expressed as a function of H_0, r_d and k_t

$$I_t = \frac{H_o}{3600} \cdot r_d \cdot k_t \quad [kW/m^2] \tag{B.2}$$

where r_d is ratio, diffuse radiation in hour/diffuse in day.

In literature, there are several different correlations of the diffuse fraction with the clearness index k_t, obtained from experimental data gathered in different sites in many years. In general, it is possible to approximate this correlation with a piecewise linear function, the correlation between the daily quantities holds as well for the corresponding hourly quantities (Orgill and Hollands, 1977). In particular in the typical range of variation for the k_t the correlation can be represented by the following expression:

$$k = p - q k_t \tag{B.3}$$

Substituting (B.2) and (B.3) in (B.1), I_β assumes the following:

$$\begin{aligned} I_\beta &= \left[\left(R_b + \rho \cdot \frac{1-\cos\beta}{2}\right) + \left(\frac{1+\cos\beta}{2} - R_b\right) \cdot p\right] \cdot r_d \cdot \frac{H_o}{3600} k_t - \left(\frac{1+\cos\beta}{2} - R_b\right) \cdot q \cdot r_d \cdot \frac{H_o}{3600} \cdot k_t^2 \\ &= T \cdot k_t - T' \cdot k_t^2 \end{aligned} \tag{B.4}$$

where

$$T = \left[\left(R_b + \rho \cdot \frac{1-\cos\beta}{2}\right) + \left(\frac{1+\cos\beta}{2} - R_b\right) \cdot p\right] \cdot r_d \cdot \frac{H_o}{3600} \tag{B.5}$$

$$T' = \left(\frac{1+\cos\beta}{2} - R_b\right) \cdot q \cdot r_d \cdot \frac{H_o}{3600} \tag{B.6}$$

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